R37-03: directional optical/Hubble sensitivity

Owner: HTT inference with retained obsstat features. Claim tier: conditional released-observable sensitivity and finite synthetic method control. Artifact mode: diagnostic results for P3/P4. Transfer source: none. Allowed use: source-qualified apparent angular magnitude/Hubble ratios and explicitly assumed physical intervals. Forbidden use: absolute H0, observed curvature, matter-congruence shear, Bianchi family, native solver, or independent SN/CF4 joint inference. Null status: conditional fixed-design Gaussian coefficient event; empirical null coverage unknown. Covariance status: the validated selected released STAT+SYS subblock is consumed once. The original 42-node acceptance stays **17/42**; the separate analytic alternate stays **18/42**.

The primary Pantheon+SH0ES retained fit has 490 selected rows (342 low, 148 high), including 62 duplicate-CID rows, and a validated full 16×16 coefficient covariance. The overlapping 705-row lower-cut sensitivity has 557 low and the same 148 high rows, including 114 duplicate-CID rows. The full 1,701-row covariance failed its source symmetry gate outside the selected union and is not used. The released chart is RA/DEC Cartesian; its physical sky-frame interpretation is unknown. The fit uses `zHD` in the dimensionless distance integral and `zHEL` in the luminosity factor. A fitted magnitude intercept and high-bin level preclude an absolute H0 estimate.

At the declared directions x=(1,0,0) and reference z=(0,0,1), the primary low-bin magnitude difference is -0.022814 ± 0.038735 mag (marginal one standard deviation). The signed dipole and full STF tensor contribute +0.000417 and +0.010089 to apparent $\Delta \ln H$ respectively. Pushing the **one simultaneous 16-dimensional 95% ellipsoid**, with squared radius 26.296228 under the stipulated Gaussian law, gives apparent H_F(x)/H_F(z) in **[0.9222, 1.1074]**. With declared $|\Delta r_g| \le 0.01$ and $|\Delta \tau| \le 0.02$, the conditional physical local-Hubble ratio region is **[0.9040, 1.1297]**. These error bounds are sensitivity inputs, not measured source-response or opacity constraints. Rows for x, y, −x; both bins; four budgets; and the 705-row sensitivity appear in `LOCAL_H_IDENTIFIED_REGION.csv`. `signed_tensor_matrices.csv` preserves all five signed STF components rather than only their norm.

The high-minus-low x-versus-z magnitude contrast is +0.097139 mag with joint standard error 0.077377 mag. Dropping the cross-bin covariance would yield 0.077829 mag. This is a falsifying comparison, not a second independent fit. `joint_depth_contrasts.csv` retains

the correlated contrast, including the overlapping 705-row sensitivity. A scalar opacity-aware baseline that applies the same linear functional to the same 16-vector and covariance has identical support to the tensor-coordinate route; `same_input_scalar_comparator.csv` records the matched mean and joint standard error for both cohorts. Both routes apply the same additive `Δr_g+Δτ/2` physical budget. The tensor representation preserves signed directional interpretation; it creates no extra information over correctly specified joint GLS.

The separately qualified CF4 fixed-weight operator supplies apparent H_W at four `z_CMB` window centers. For x it ranges from 72.59 to 72.65 km s^{-1} Mpc^{-1} in the first two windows and differs from the z direction. Its retained 36×36 covariance is applied to each displayed marginal standard error in `CF4_retained_H_windows.csv`. CF4 and the released SN cohort have different sources, nuisance laws, redshift charts and unknown shared peculiar-velocity covariance. No joint significance or absolute comparison is inferred.

For the exact finite synthetic CBL control, set flat de Sitter `H=70 km s⁻¹ Mpc⁻¹`, `d=cz/H`, and transparent-at-vertex `τ(z)=0.2z`. At the four CF4 window centers z=0.025, 0.04, 0.06, 0.08, the naive first-derivative curvature `(H²d_r′²/c²−1)/d_r²` is positive, from 8.718×10^{-7} to 2.721×10^{-7} Mpc^{-2}. Correcting with the supplied same opacity gives zero to 2×10^{-20} Mpc^{-2} and a common zero constant across windows. The finite family `τ=az`, `a=0,…,0.25` in steps of 0.001 contains this control; its pointwise compatible intervals include zero. The same `a` is held across windows for the common-constant test. `synthetic_CBL_first_derivative.csv` records each value. This establishes only an algebraic/finite-method control: the angular SN carrier has no radial distance derivative, and there is no overlapping independently qualified H(z) and joint law for an empirical CBL curvature result.

For optical transmission, the exact synthetic filter `K=diag(1/3,1/5)` gives `Q_in=diag(1/9,1/25)`. Trace-nonincreasing alone gives `0<D_L≤3` in the example length unit. External calibration `t_min I≤T≤I` gives `sqrt(t_min/(1/25))≤D_L≤3`; at `t_min=0.1` this is [1.58114,3]. `distance_transmission_interval.csv` and its vector PDF show the finite region. A traceless matrix log fixes relative channel transmission and dichroic shape, while grey transmission remains a separate scalar. A constant angular calibration/foreground offset can mimic the local slope; transparent-at-vertex propagation requires τ(0,n)=0. Photon redshift, source chart and observer boost need a joint response before physical interpretation.

All figures are vector PDFs generated by `compute_hubble.py`, which reads only saved R36 results. `local_H_region_vs_opacity.pdf` uses the primary selected SN law; `distance_transmission_identified_`

`interval.pdf` is synthetic optical tomography. Both display conditional regions, not posterior intervals or observed curvature. Assumptions use signature (−,+,+,+), explicit c in the synthetic curvature units, outward chart directions, Frobenius-normalized STF basis, and the retained source redshift conventions. The FLRW/no-opacity limit and equal-direction contrast are checked; rank/denominator failures near `d=0` are excluded from the finite CBL windows. No catalogue replay, native solver, or ODE was run.